

Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. • **v2.3** • 15 August 2026 • supersedes the v2.2 PDF of the same date

What this revision does. v2.2 named A2 and A4 as the whole of the critical path and put the honest remaining total at ~43.75 → 41.75 engineering days. Session 6 built both. **The application now starts.** It reads its configuration, opens a database, resolves which tenant a message belongs to, classifies it, and files the decision — end to end, through the production wiring, for the first time. What it still cannot do is reply: that is A5 and A6, and those two items are now the whole of Track A. **Remaining: ~40.25 engineering days.**

Where we stand today

Built and tested	Missing — nothing ships without these
375 passing tests, no DB or network needed 83 source files across 20 modules (121 with tests) 19 DB tables + 3 migrations Every external system behind a swappable seam Eval runner + CI accuracy gate, 50 golden cases Receptionist vocabulary: 19 intents, data not code Prompt v3 — catalogue rendered from the vocabulary Channel-neutral envelope + chat/voice adapters Task state machine and the autonomy gate Typed config over every .env variable (A1) Anthropic + OpenAI on the LLM seam, caching on (A3) NEW • Engine + session scope, pooler-safe by default (A2) NEW • Composition root: the process starts and files a message end to end (A4)	No DB connection — done (A2) No endpoint — done (A4) No outbound — ChannelSender has no implementation No memory across turns — worker.py still passes task=None No Dockerfile — so cd.yml has never deployed anything No Supabase project — and no channel_configs row, without which every webhook 500s (B1) No RLS — no policy in any migration No control page — one CSS file, no package.json No control-page API — ~25 endpoints, none written Knowledge — zero tables, zero rows Live availability — no read path to any PMS A measured prompt — the gate replays v2 fixtures, so 88% is not v3's number

What changed in this revision (v2.2 → v2.3)

Delivered	Consequences
A2 and A4 delivered — 1.5 engineering days. Total 41.75 → 40.25 Track A: 4.75 → 3.25 days, and it is now only A5 and A6 Tests 325 → 375 (+50). Source files 79 → 83 Five database port implementations replace the last of the in-memory doubles Two runtime dependencies: psycpg 3 and uvicorn	DATABASE_POOL_MODE is new, and pooler-safe by default — the Supabase pgbouncer risk is now a policy rather than an open question Migrations connect exactly as the app does, so B1 cannot discover a driver mismatch on its first upgrade Tenant resolution refuses to guess — B1 must insert a channel_configs row This generator is regenerated, so the three-disagreeing-artifacts item is closed Unchanged: 2.4 is still the last item before a demo; 3.1 still blocked on P1; the scope guard still holds — no PDF handbook ingestion

Totals: ~40.25 engineering days remaining. At the observed 2–3 engineering days per working session that is **13–20 sessions**; at the six-day week these estimates assume, ~6.7 weeks of pure build — so plan **~9 weeks to sellable** and **~10 days to a real number that answers safely**. The critical path is still Track A, and it is now A5 then A6.

Track A — Make it run. [FOUR OF SIX DONE — 3.25 DAYS LEFT]

The bottom four layers exist. What remains is the two items that turn filing into answering, and they are the last thing between this and a receptionist.

#	Work item	Urgency	Ease	Days	Status
A1	Configuration layer	DONE	Easy	0.5	DONE — core/config.py; every variable in .env.example typed, required per subsystem, placeholders read as absence, secrets as SecretStr
A2	DB engine and session	DONE	Easy	0.5	DONE — db/engine.py; engine, sessionmaker, session scope. Transaction-mode pooling assumed by default (NullPool + prepare_threshold=None), DATABASE_POOL_MODE opts out; a bare postgresql:// URI is rewritten onto psycopg 3; alembic connects the same way
A3	Concrete LLM providers	DONE	Moderate	1.5	DONE — Anthropic + OpenAI on the existing seam; system block marked cacheable; per-call usage reported, so cost is measurable
A4	Composition root	DONE	Moderate	1.0	DONE — apps/api/main.py, run as uvicorn apps.api.main:create_application --factory. Also the five database port implementations create_app needed and a process-level worker pool, so the webhook still answers before classification runs
A5	Wire continuity + excise v1 filer	NOW	Moderate	2.25	NEXT — ConversationRepository is still never called outside tests; the orchestrator passes task=None and empty slots. Also converts process() to async and removes the rules/destinations threading (D24). Tables retained, not dropped
A6	Outbound sender	NOW	Moderate	1.0	NOT BUILT — ChannelSender has no implementation; replies are composed and never sent. Also the moment to move the channel credential fields out of core/config.py and clear the last KNOWN_LEAKS entry

Why A5 is the whole game now. Items 2.1 and 2.2 are genuinely written — the tables, the repository, the task state machine, the reply path. What was never written is the line joining them, and A4 deliberately did not write it: wiring a receptionist before continuity exists produces one that forgets the previous turn, and it would have nothing to send with until A6. The milestone “holds a conversation” is still not true, and A5 is what makes it true.

Track B — Host it. [Supabase + Render]

Stack locked 15 August 2026. B2 must land before a second client exists. B1 grew one precondition this session — see its note.

#	Work item	Urgency	Ease	Days	Status
B1	Supabase project and migrations	NOW	Easy	0.5	NOT STARTED — provision, set DATABASE_URL, run the 3 existing migrations. And insert one channel_configs row per endpoint: tenant resolution reads that table and raises rather than guessing, so without the row every inbound message 500s
B2	Row-Level Security	NOW	Moderate	1.5	NOT BUILT — no policy in any migration, though AGENTS.md calls tenant isolation non-negotiable. Per-session GUC + a cross-tenant read test
B3	Dockerfile and Render service	HIGH	Easy	0.75	NOT BUILT — the Dockerfile alone activates the dormant cd.yml image job. The start command is now a known quantity (uvicorn --factory)
B4	Domain, TLS, webhook subscription	HIGH	Easy	0.5	NOT STARTED — the platform needs a public HTTPS URL before anything arrives
B5	Durable queue (Redis + arq)	MED	Moderate	1.0	NOT BUILT — the in-process worker pool A4 ships loses in-flight classifications on every deploy. Persist-before-enqueue means a message is never lost, only its classification. Same seam, same consumer

Runs in parallel — start today. P1 pick the first client (NOW, blocks 3.1) • P2 PMS sandbox keys (HIGH, 0.5d) • P3 file the platform business verification (HIGH, 1 hr) • P4 Graphify as a build aid (LOW, optional). Founder time, excluded from the total — and P2 and P3 are the two that make Track B's week land on time.

Track 2 — Make it a receptionist. [2.4, 2.7, 2.8 REMAIN]

What remains is knowledge, a measured prompt, and more than one property.

#	Work item	Urgency	Ease	Days	Status
2.1	Conversations, tasks and slot filling	HIGH	Hard	spent	PART-BUILT — 2.0d spent. The wiring is priced in A5, not here
2.2	The reply path	HIGH	Moderate	spent	PART-BUILT — 1.5d spent. The sender is priced in A6, not here
2.3	Autonomy gate	DONE	Easy	1.0	DONE — RECEPTIONIST_REPLY as the fourth RoutingAction
2.4	Knowledge base	HIGH	Moderate	2.0	NEXT after Track A — facts table with sensitivity flags, a real “I don’t know” that fetches a human, verification codes
2.5	Prompt v2 and rewrite the golden set	DONE	Moderate	1.0	DONE — golden set 8 → 50 across all 19 intents
2.6	Intent taxonomy unification	DONE	Easy	0.5	DONE — 19 vocabulary-aligned intents
2.7	Re-record the eval under prompt v3	MED	Easy	0.5	UNBLOCKED — A3 landed; needs only a key. Add Franco-Arabic cases, and settle the two open numbers on the next page
2.8	Many properties per client	HIGH	Moderate	1.0	properties table, per-property fact scoping, message → property resolution

The eval is still not measuring the prompt. The gate runs and passes, but it replays fixtures keyed by message text and recorded under prompt v2, so it reports 88% whatever the prompt says. Until 2.7 runs, treat 0.88 as v2’s number. baseline.json says so in the file, and correctly records claude-haiku-4-5-20251001 as the model that produced those fixtures.

Tracks D, G, E and 3 — unchanged from v2.1

Summarised here; the v2.1 PDF carries the full per-item detail.

D — The control page (D0–D10) 13.75d	Six receptionist views. D2 (REST API, 3.0d) is the hidden half — “build the control page” reads as frontend work; three of its days are backend endpoints that do not exist. Still the item most likely to be cut by accident
G — Receptionist guardrails (G1–G4) 5.5d	Not new features — these enforce rules intents.yaml states and schema.py validates and nothing in apps/ honours. G3 (emergency path, 1.5d) is not optional before a real guest: worker.py hardcodes emergency=False, so a gas leak files a maintenance ticket
E — What makes it sellable (E1–E5) 4.5d	Tenant onboarding, usage metering, billing, observability, backups and residency
3 — Integration and measurement (3.1–3.3) 5.5d	3.1 blocked on P1. Base360.ai ruled out; Hostaway, Guesty and Cloudbeds all publish APIs

New this revision: what A2 and A4 settled, and what they left open

The connection path	Supabase's application URI is pgbouncer in transaction mode : a client connection is bound to a server connection for one transaction only. Neither psycopg's server-side prepared statements nor SQLAlchemy's own connection pool survives that, and both fail intermittently under load rather than in a test. DATABASE_POOL_MODE defaults to the pooler-safe policy (NullPool + prepare_threshold=None), which is always correct and merely slower when unnecessary; session opts out for a direct connection. It is a variable rather than a guess from the port because a pooler can sit in front of any host. The load test the last three revisions asked for is now a flip of that variable, not a code change
The driver	A bare postgresql:// URI resolves to psycopg 2, which this project does not ship. It is rewritten onto psycopg 3 so the URI can be pasted from the dashboard — and alembic/env.py now goes through the same two functions, because migrations resolving a different driver from the application is something a deploy discovers on the day it matters
Tenant attribution	An inbound message is attributed through channel_configs , and an endpoint with no enabled row raises rather than falling back to a default tenant — guessing writes one customer's message into another's account. The lookup does not filter by channel kind, so a phone line needs no code change. B1's checklist grew one line because of this
Still not written	classifications rows . The audit row carries the classification snapshot, but that table wants latency_ms and prompt_version and the orchestrator surfaces neither; model_used has nowhere to go either. Small, and it belongs with A5, which is already in that code path. Do not invent the missing telemetry to fill the columns

Two numbers nobody has measured, both cheap to settle during 2.7

1. The prompt's real token count	~5k is a characters÷4 estimate. Haiku 4.5 will not cache a prefix below 4,096 tokens , and the cheap tier is where caching pays for itself. The estimate clears the floor by ~30%, and the true count is probably higher (Arabic and Franco-Arabic tokenize worse than the approximation), so caching almost certainly activates — but if the vocabulary shrinks it stops silently: no error, just a larger bill. Sonnet 5's floor is 1,024, so escalation is not exposed either way
2. Cost per message	Still unmeasured, but measurable : TokenUsage reports cached versus fresh input tokens per call, normalised across both providers. Measure before quoting a price

How the total reconciles

A 3.25 + B 4.25 + D 13.75 + E 4.5 + G 5.5 + (2.4, 2.7, 2.8) 3.5 + (3.1, 3.2, 3.3) 5.5 = **40.25**

Rows 2.1 and 2.2 show "spent" rather than a number because their days are already delivered and the outstanding wiring is priced in A5 and A6 — adding them would double-count 3.5 days. P1–P4 is excluded: founder time, not engineering. Counting scope loosely is what produced the 5.75 figure that v2.1 replaced, so the arithmetic is stated rather than implied.

The dates

Milestone	v2.2 said	Now	Status
M1 — answers a real message, safely	~11.5 days	~10 days	NEXT — Track A (3.25 left) + B1–B4 + G3 + 2.4. The emergency path is not optional here
M2 — you can watch and correct it	+13.75 days	+13.75 days	PENDING — Track D, six receptionist views
M3 — sellable	+16.5 days	+16.5 days	PENDING — Tracks E, F and the rest of G. Blocked on P1
Phone answering	after M3	after M3	PENDING — the speech-to-text seam already exists

What would actually move these dates

The risks v2.2 named, re-scored against what session 6 learned.

Supabase connection pooling	Downgraded from imminent to unmeasured. A2 assumes the transaction pooler and is safe there by default. What is not known is the cost of that safety — NullPool trades a handshake per checkout for correctness. Measure it during B1/B3 and flip DATABASE_POOL_MODE if session mode wins; do not settle it in production at 2am
A missing channel_configs row	NEW. The first deployed webhook will 500 on every message until B1 inserts one row per endpoint. It is a one-line fix and a genuinely confusing hour if nobody wrote it down, which is why it is in B1's status, the README and the spec
RLS retrofitted late	Unchanged. Adding row-level security after application queries exist surfaces as silent empty result sets rather than errors. B2 sits before the second tenant for that reason
D2 is invisible in its own track title	Unchanged, and still the most likely accidental cut. Three of Track D's days are backend
Cost per message	Unmeasured, not unknown. A3 made it measurable and turned prompt caching on. Measure before quoting a price
PMS API access	Unchanged. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 today
Three roadmap artifacts disagree	CLOSED. docs/make_roadmap.py now generates this document into docs/Watcher_v2_Roadmap.pdf. Edit the generator and re-run it; do not hand-write a fourth version

If you do only one thing next session: A5, then A6. Not 2.4. The receptionist can now read its configuration, reach a model, connect to a database and start — and a guest who messages it gets silence, because nothing joins the conversation to the task and nothing puts a reply on the wire. A5 + A6 is 3.25 days and it is the difference between a system that files and a receptionist — and G3 before any real guest can message it.

Session log

Session	Delivered	Tests	Days
1	PR #13 — items 1.1, 1.3, 2.1, 2.2, 2.3	228 → 248	6.5
2	PR #14 and #15 — items 2.5, 2.6	248 → 259	1.5
3	PR #16 — prompt v3, unplanned	259 → 277	~0.5
4	Code audit plus a design review. No application code changed. Five unpriced tracks added; 2.1/2.2 downgraded to part-built; emergency detection found unwired; the control page rebuilt around six receptionist views. Decisions D14–D26	277	—
5	PR #17 — items A1, A3. Config layer over all ~20 env vars; Anthropic + OpenAI providers with prompt caching and per-call usage; D8-a re-pinned to the Claude 5 family	277 → 325	2.0
6	Items A2, A4. Engine and session scope, pooler-safe by default; the composition root and the five database port implementations it needed; a process-level worker pool so the webhook still answers before classification; alembic aligned to the same connection path; this generator regenerated	325 → 375	1.5

Cumulative: 12.0 engineering days delivered. ~40.25 remaining.